

SHRI RAM MURTI SMARAK
COLLEGE OF ENGINEERING TECHNOLOGY AND
RESEARCH, BAREILLY



Project Topic:

AI Resume Shortlister

Presented by:

Raj Singh

2304500100083

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Problem Statement

The Challenge

- Companies and recruiters still evaluate candidates manually by reading resumes one by one and comparing them against job descriptions — this takes significant time and often leads to inconsistent hiring decisions and missed qualified candidates.
- Candidate information and skill requirements are scattered across multiple resume formats (PDF, DOCX, TXT) and job description documents, making it difficult for hiring teams to quickly identify top matches and create accurate evaluation reports.
- Small and medium-sized companies usually don't have access to expensive enterprise recruiting software where candidate evaluations, scoring, and team feedback can be managed in one centralized location.

Problem Statement

Our Solution

- TalentAI is an AI-powered recruitment platform that automates resume parsing, skill matching, candidate scoring, and shortlisting.
- It uses hybrid NLP techniques, combining semantic embeddings and keyword matching to evaluate candidates against job requirements.
- It provides AI-generated insights, analytics dashboards, candidate evaluation tools, and PDF reports to support faster and data-driven hiring decisions.

About The Internship

- **Organization:** SkillInfyTech IT Solutions Private Limited
- **Internship Type:** Technical Internship
- **Domain:** Artificial Intelligence
- **Duration:** 07 June 2026 – 16 July 2026
- **Role:** Artificial Intelligence Intern
- Gained practical experience through **AI training, practical assignments, and project-based learning.**
- Participated in **training sessions and hands-on technical activities** related to AI
- Developed skills in **problem-solving, technical learning, and project development.**
- Gained exposure to **real-world AI applications and industry-oriented practices.**
- Improved **professionalism, teamwork, discipline, and technical capabilities** through mentorship and practical tasks.

Company Introduction

- SkillInfyTech IT Solutions Private Limited is an IT and software technology company.
- The company was established in 2024 with a focus on technology and skill development.
- It provides IT solutions, professional training, and internship programs.
- The company focuses on industry-oriented and practical learning.
- It offers exposure to AI, ML, Data Science, Full Stack Development, Cloud Computing, and Cybersecurity.
- It works with students, freshers, professionals, and organizations.
- It helps bridge the gap between academic education and industry requirements.
- It provides opportunities to gain hands-on projects and real-world technical experience.

Internship Objectives & Learning Goals

1. To gain **practical knowledge of Artificial Intelligence** in a real-world development environment.
2. To understand **AI concepts, techniques, and their practical applications**.
3. To develop skills in **machine learning, data processing, and AI-based problem solving**.
4. To apply AI concepts through **practical assignments and project-based tasks**.
5. To improve **problem-solving, analytical, and technical skills**.
6. To gain exposure to **industry-oriented practices, teamwork, mentorship, and project workflows**.

Project Introduction

- **Project Name:** TalentAI
- **Project Type:** AI-Powered Resume Screening & Candidate Ranking System
- **Purpose:** To automate **resume screening, candidate evaluation, and shortlisting.**
- Provides a centralized platform for **resume analysis and candidate matching.**
- Uses **NLP, semantic embeddings, and keyword matching** to evaluate candidates.
- Developed using **Python, FastAPI, Sentence-Transformers, and Scikit-learn.**
- Provides **candidate scoring, AI insights, dashboards, and reports.**
- Focuses on **faster, accurate, and data-driven recruitment.**

Project Objectives

1. To automate resume screening and candidate shortlisting using AI and NLP.
2. To accurately match candidate skills with job requirements beyond simple keyword matching.
3. To rank candidates based on their skills, qualifications, and experience.
4. To support data-driven hiring decisions through dashboards and AI-generated insights.
5. To reduce recruitment time and operational costs through automated candidate evaluation.
6. To ensure candidate privacy, fairness, and secure handling of recruitment data.

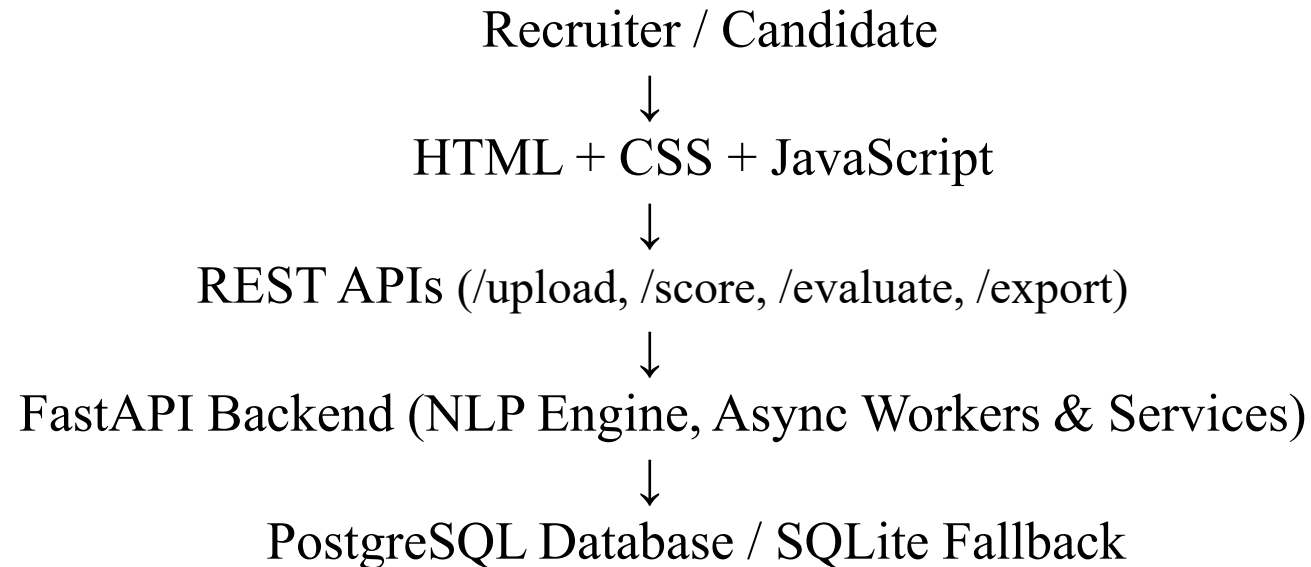
Technologies & Tools Used

- **Python** – Core programming language for AI/ML development and backend implementation.
- **Artificial Intelligence & Machine Learning** – Used for intelligent resume analysis, candidate evaluation, and ranking.
- **Sentence-Transformers** – Used for generating semantic embeddings and understanding skill similarity.
- **Scikit-learn** – Used for **TF-IDF vectorization and cosine similarity** for keyword-based matching.
- **Natural Language Processing (NLP)** – Used for resume text processing, skill extraction, and semantic analysis.
- **NumPy** – Used for numerical operations and data processing through ML libraries.
- **FastAPI** – Used to develop backend services and AI-powered APIs.
- **Git & GitHub** – Used for version control and source-code management.
- **VS Code** – Used as the primary development environment.

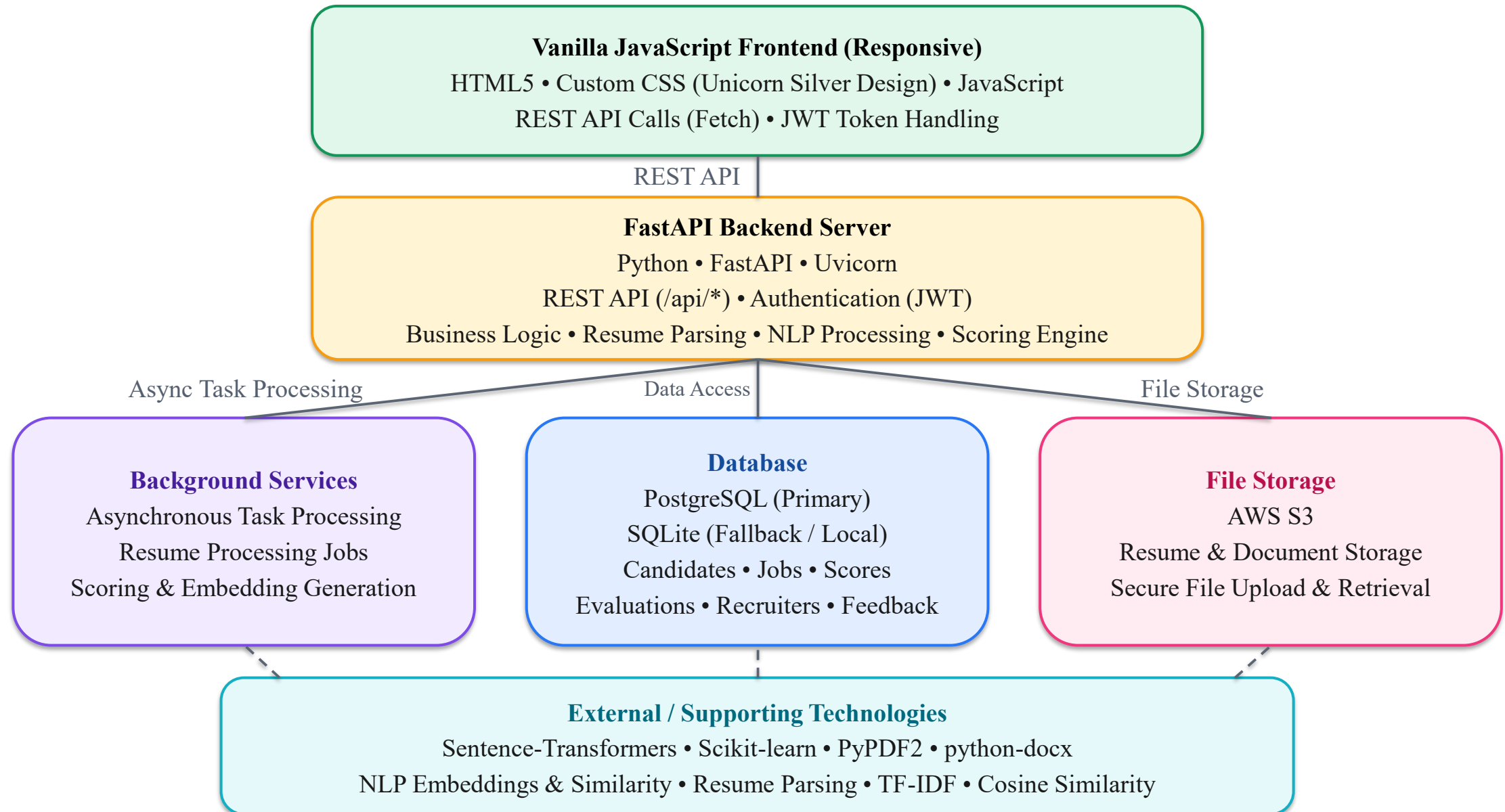
System / Project Architecture

Candidate/Recruiter → Frontend → REST API → FastAPI Backend → PostgreSQL Database

Architecture Diagram



System Architecture



Implementation & Working Process

- **User Authentication & Upload** – Recruiter logs in and uploads **resumes and job descriptions** through the frontend.
- **Document Processing** – FastAPI processes **PDF, DOCX, and TXT** files and extracts relevant candidate information.
- **NLP & Skill Matching** – The system uses **Sentence-Transformers and TF-IDF** to extract skills and match candidates with job requirements.
- **Scoring & AI Evaluation** – Candidates are **scored and ranked**, with AI-generated verdicts, pros/cons, and interview questions.
- **Results & Decision** – Scores and analytics are stored in the database and displayed through **dashboards and reports** for recruiter decision-making.

Working Flow

Recruiter → Login → Frontend → Resume & Job Description Upload → REST API → FastAPI → Document Processing → NLP & Skill Matching → Score & Ranking → AI Evaluation → Database → Analytics & Display → Recruiter Decision

Database

- **Database:** PostgreSQL (with SQLite fallback).
- PostgreSQL is used as the primary database, with **SQLite automatically used when PostgreSQL is unavailable.**
- The database manages **candidate data, evaluation information, scores, recruiter notes, and candidate statuses.**
- **Alembic** is used for database migrations and table initialization.

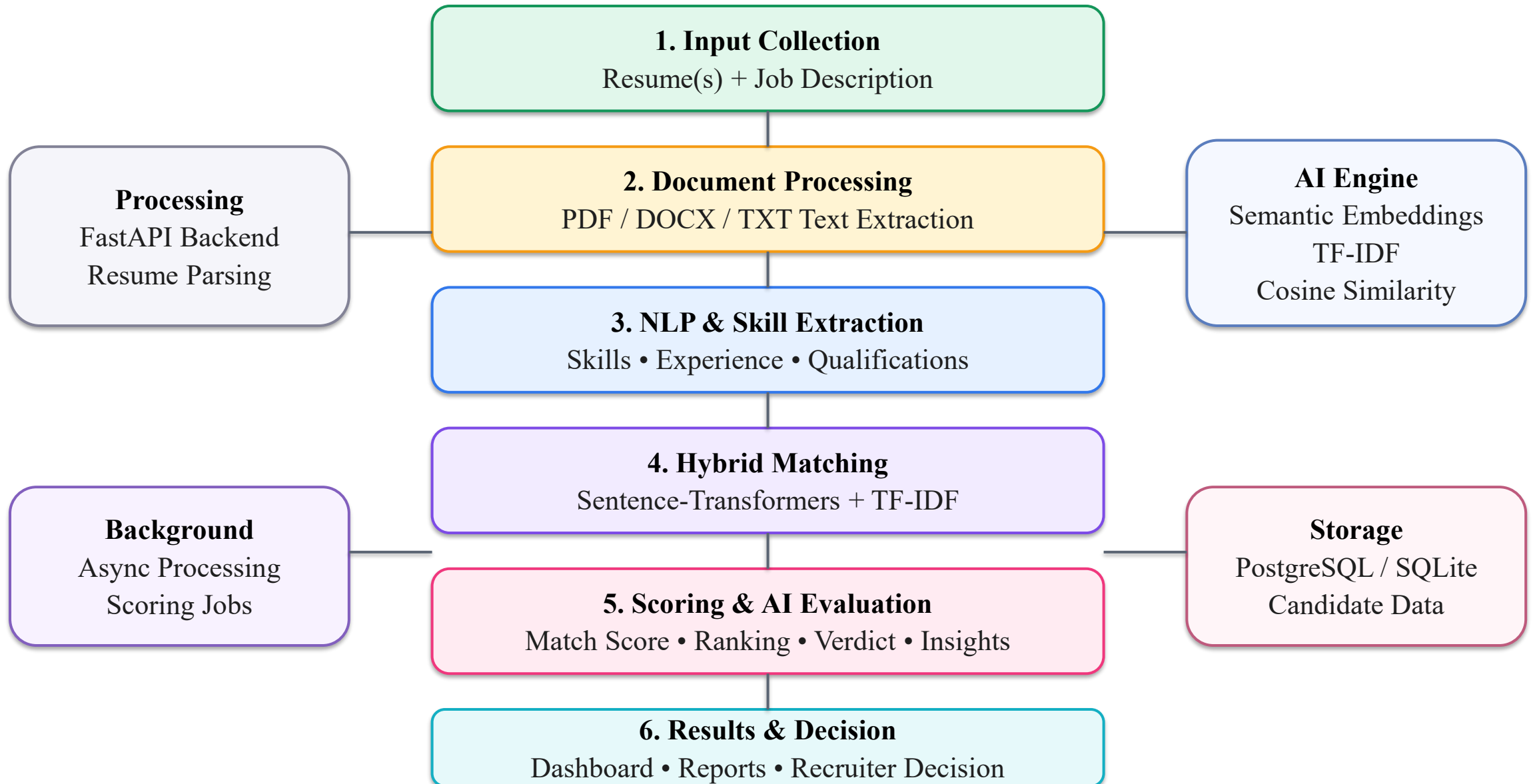
API(Application Programming Interface)

- **REST APIs** are used for communication between the frontend and FastAPI backend.
- **FastAPI** is used to build the backend server and REST endpoints.
- The backend handles **resume processing, candidate scoring, evaluation, and related recruitment operations.**

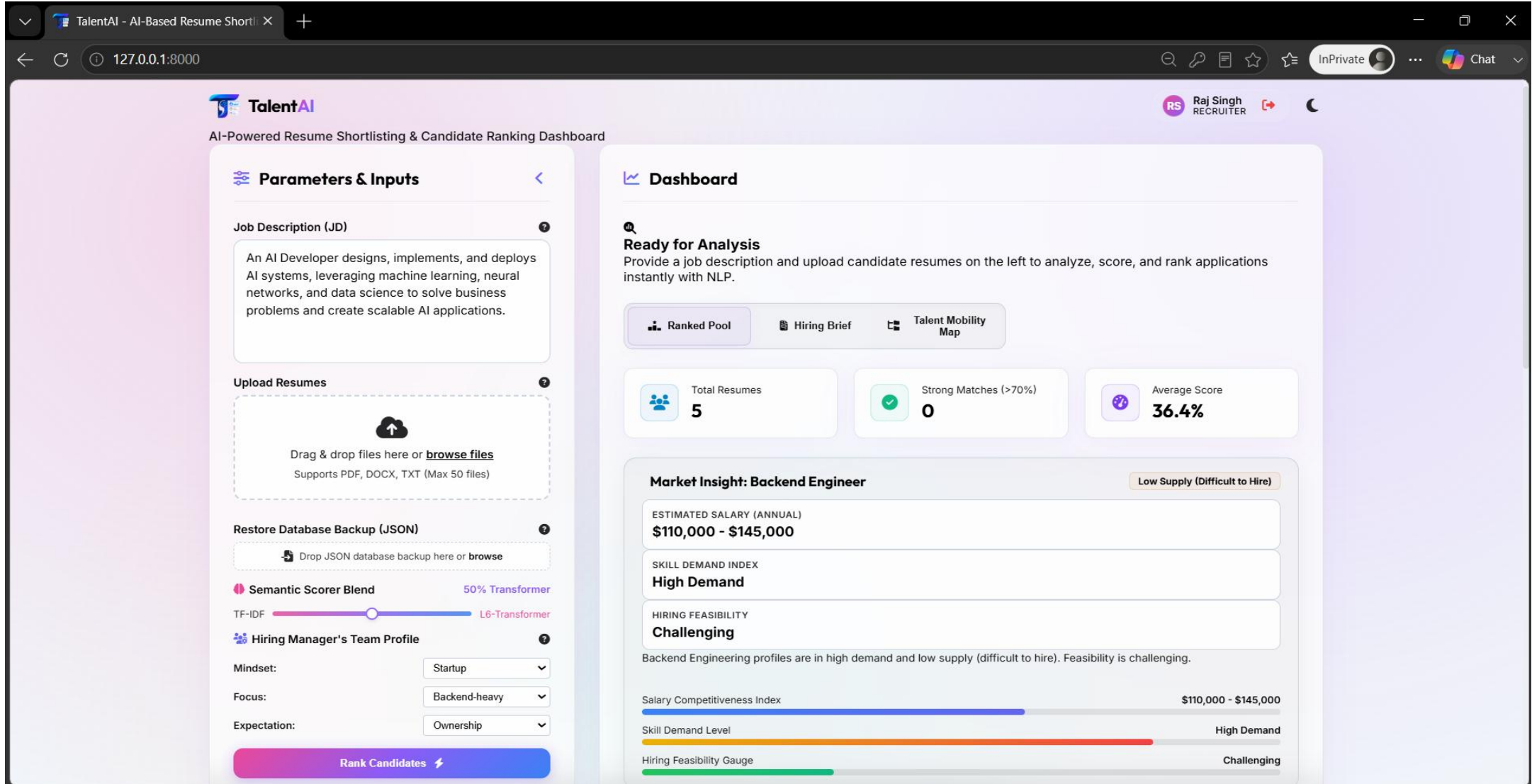
Technology Stack

Stack	Technology	Purpose
Frontend	HTML5 + CSS + JavaScript	User interface and interaction
Backend	Python + FastAPI	Backend services and REST APIs
NLP / AI	Sentence-Transformers	Semantic embeddings and skill matching
Machine Learning	Scikit-learn	TF-IDF and cosine similarity
Database	PostgreSQL / SQLite	Candidate, job, score and evaluation data
Document Processing	PyPDF2 + python-docx	Resume text extraction
Async Processing	Celery + Redis	Background task processing
File Storage	AWS S3	Resume and document storage
Development & Version Control	Git, GitHub + VS Code	Code management and development

Project Methodology



Results & Snapshots



TalentAI
AI-Powered Resume Shortlisting & Candidate Ranking Dashboard

Parameters & Inputs

Job Description (JD)
An AI Developer designs, implements, and deploys AI systems, leveraging machine learning, neural networks, and data science to solve business problems and create scalable AI applications.

Upload Resumes
Drag & drop files here or [browse files](#)
Supports PDF, DOCX, TXT (Max 50 files)

Restore Database Backup (JSON)
Drop JSON database backup here or [browse](#)

Semantic Scorer Blend
50% Transformer
TF-IDF L6-Transformer

Hiring Manager's Team Profile

Mindset:

Focus:

Expectation:

Rank Candidates

Dashboard

Ready for Analysis
Provide a job description and upload candidate resumes on the left to analyze, score, and rank applications instantly with NLP.

Ranked Pool **Hiring Brief** **Talent Mobility Map**

Total Resumes
5

Strong Matches (>70%)
0

Average Score
36.4%

Market Insight: Backend Engineer **Low Supply (Difficult to Hire)**

ESTIMATED SALARY (ANNUAL)
\$110,000 - \$145,000

SKILL DEMAND INDEX
High Demand

HIRING FEASIBILITY
Challenging

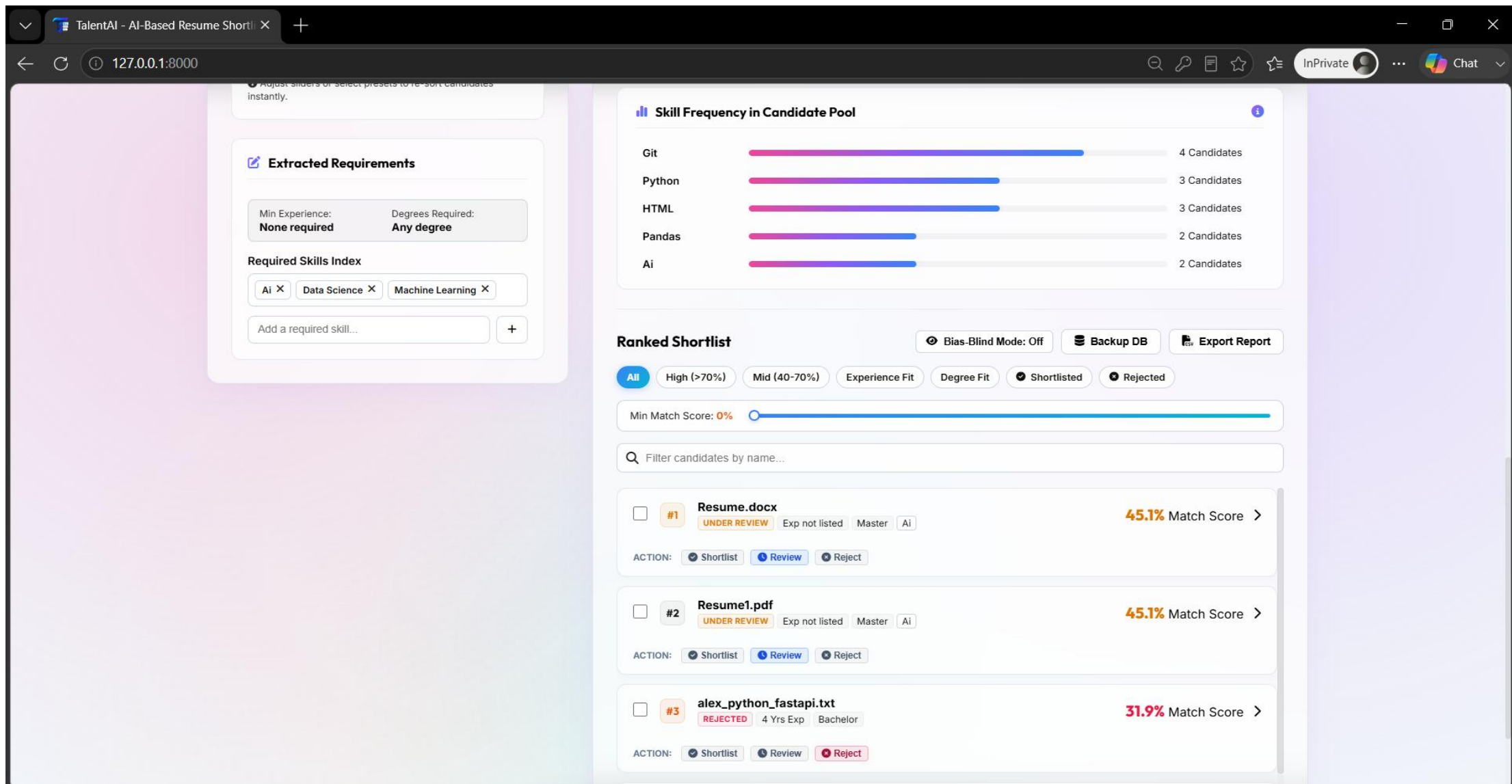
Backend Engineering profiles are in high demand and low supply (difficult to hire). Feasibility is challenging.

Salary Competitiveness Index
\$110,000 - \$145,000

Skill Demand Level
High Demand

Hiring Feasibility Gauge
Challenging

Application Preview: Dashboard



Resume Score

TalentAI - AI-Based Resume Shortl...

127.0.0.1:8000

InPrivate

Chat

#1 Resume.docx

PDF Report

EVALUATION STATUS:

Shortlisted

Under Review

Rejected

45.1% Match

Semantic Relevance (TF-IDF Cosine) 21.1%

Required Skills Coverage 33.3%

Experience Alignment 100.0%

Explainable AI Score Breakdown

Domain Fit 100%

Seniority Fit 100%

Culture & Soft-Signal Fit 50%

Hiring Manager Team Fit 73%

Why This Candidate?

Good Match: This candidate fits the core requirements with 0.0 years of experience. They cover skills like Ai. Additionally, they show high compatibility with the team's AI / Machine Learning focus, demonstrating a Enterprise mindset.

Job-to-Team Fit Dimensions

Mindset Alignment: Enterprise (Partial Match)

Resume Report

🔧 AI Candidate Fit Verdict

Under Review: *Resume.docx* is a partial match with a **45.1% Match Score**. They have **0 years** of experience and match **1 of 3** required skills. They exhibit missing required skills in key JD areas: *Data Science, Machine Learning*. Recommended next step: Conduct a phone screening call to assess their competency details in the missing core skills.

v2.1.0

✔ **MATCH INDICATORS**

⚠ **OPTIMIZATION FLAGS**

👤 Soft Skills & Professional Traits

No leadership or architectural soft traits flagged.

Recruiter Comments & Notes

Write internal screening notes, candidate impressions, or feedback here... (saves automatically)

Pros & Cons Strength Analysis

+ Candidate Strengths (Pros)

- Strong core matches for required skill chips: *AI*.

- Gaps & Development Areas (Cons)

- Gaps identified in required skills: *Data Science, Machine Learning*.

📁 Interactive Interview Kit

📌 Role-specific round-based sets matching candidate gaps.

Screening

Technical

System Design

Behavioral

What elements of our team's Enterprise mindset align with your career goals, and why do you want to join us?

📄 Copy

You mention experience in Ai. Can you quickly summarize a recent project where you applied these?

📄 Copy

📄 Full Parsed Resume Text

📌 On-demand text preview of the candidate's parsed resume file.

🔍 Hide Resume Text

Raj Singh
Email: rajsingh98447859@gmail.com | Phone: 7388445597
LinkedIn: www.linkedin.com/in/raj-singh-49aa842a5

Objective
Motivated and enthusiastic Computer Science student seeking an internship opportunity to apply my programming skills and gain hands-on experience in software development.

Education
Shri Ram Murti Smarak College of Engineering Technology and Research, Bareilly, Uttar Pradesh
B. Tech in Computer Science and Engineering (Expected: 2027)

Technical Skills
Programming Languages: Python, HTML
Frameworks & Libraries: Streamlit, Tkinter, Pandas, NumPy, Matplotlib

Profile Matching Details

📄 Experience Profile

📄 Education Profile

Profile Matching Details

📄 Experience Profile

Required: **0 Years (None)**

Found: **0 Years**



📄 Education Profile

Degrees Found: **Master**

🚩 **Low Confidence**

Requirements Match:

Matched

JD Matching Summary

✔ Matched Required Skills

AI

⚠ Missing Required Skills

Data Science

Machine Learning

All Extracted Skills (NER)

LANGUAGES

HTML

Python

FRAMEWORKS & LIBRARIES

NumPy

Pandas

CLOUD & DEVOPS

GCP

Git

METHODOLOGIES & DOMAINS

AI

Web Development

Resume Highlights Preview

💡 Matched skills are highlighted in green, missing required skills are highlighted in red.

Complete Analysis of Resume

127.0.0.1:8000

Shortlist Comparison Grid

Side-by-side analysis of candidate alignments and qualifications.

Qualification / Metric	Candidate #1 Resume.docx	Candidate #2 Resume1.pdf	Candidate #3 alex_python_fastapi.txt
Overall Match Score	45.1%	45.1%	31.9%
Review Status	Under Review	Under Review	Rejected
Semantic Relevance	21.1%	21.2%	17.3%
Skills Coverage	33.3%	33.3%	0.0%
Experience Alignment	100.0%	100.0%	100.0%
Years of Experience	0 Years (Required: 0 Yrs)	0 Years (Required: 0 Yrs)	4 Years (Required: 0 Yrs)
Degree Extracted	Master	Master	Bachelor
Degree Match Status	✓ Matched	✓ Matched	✓ Matched
Matched Skills	Ai	Ai	None
Missing Required Skills	Data Science Machine Learning	Data Science Machine Learning	Ai Data Science Machine Learning

Comparison of Resume

Future Scope

- **Advanced AI Matching** – Improve candidate-job matching using advanced LLMs and semantic analysis.
- **AI-Powered Interviews** – Introduce automated **voice/video interviews** and candidate assessment.
- **Predictive Hiring Analytics** – Predict **job fit, candidate performance, and hiring success** using historical data.
- **Multi-Language Support** – Enable resume processing and evaluation in **multiple languages**.
- **ATS Integration** – Integrate TalentAI with existing **Applicant Tracking Systems and recruitment platforms**.
- **Cloud Scalability** – Scale the platform for **large organizations and high-volume recruitment**.

Conclusion

- **TalentAI** successfully automates the **resume screening, candidate matching, scoring, and shortlisting** process.
- The system uses **AI, NLP, semantic embeddings, and TF-IDF** to provide accurate candidate-job matching.
- It reduces **manual effort, recruitment time, and operational costs** while improving consistency in candidate evaluation.
- Features such as **AI-generated insights, candidate rankings, analytics, and evaluation reports** support data-driven hiring decisions.
- Overall, TalentAI provides a **smart, efficient, and scalable solution** for modern recruitment and can be further enhanced with advanced AI capabilities.

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- [6] “Automated Resume Parsing and Ranking using Natural Language Processing,” in *Proc. IEEE Conference*, 2024.

Thank You